UAE FOOD SUPPLY RESILIENCE

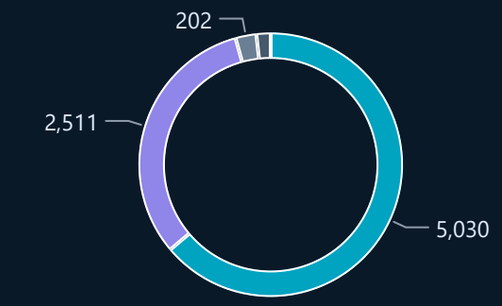
Hormuz + Red Sea/Suez stress test · baseline frozen 6 Oct 2026

Scope: C1-only | Regional envelope | C1 total (sensitivity)

Activity level: Low | Base | High synchronized

2026 demand: Low | High

2026 REQUIREMENT BY MODE · T/DAY



95.7%

Share modeled at ports

WHERE THE BASKET COMES FROM · CLICK A COUNTRY

91.8%

Port coverage



621.3

Port gap · t/day

87.65%

Container coverage

404.9

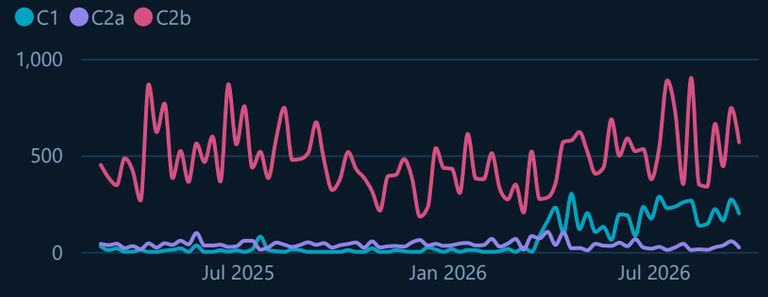
C1 total gap · t/day

MAIN SUPPLIERS · LARGEST SINGLE-PRODUCT SHARE

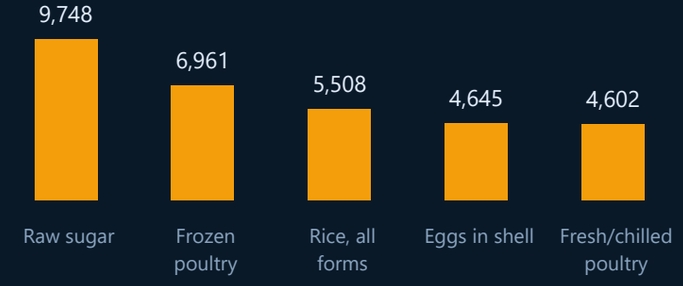
Country	Product	Share	Items
Brazil	Raw sugar	98.7%	3
India	Rice, all forms	72.1%	6
Türkiye	Eggs in shell	66.4%	1
Saudi Arabia	Fresh/chilled poultry	54.1%	8
Ukraine	Crude/upstream cooking oil	41.9%	4
Oman	Fresh/chilled poultry	40.9%	4
New Zealand	Milk powder and whey	39.6%	4
Russian Federation	Wheat grain	39.4%	4
Canada	Dry pulses	36.2%	2
Indonesia	Refined cooking oil	33.2%	1
Australia	Dry pulses	24.8%	4
Malaysia	Refined cooking oil	19.4%	2
Argentina	Crude/upstream cooking oil	18.4%	2
Poland	Packaged bread (proxy)	16.6%	1
Pakistan	Rice, all forms	16.3%	3

Items = basket items where the country is a Top-5 supplier

WEEKLY CONTAINER IMPORTS AT ALTERNATIVE PORTS · KT/WEEK



MOST CONCENTRATED ITEMS · HHI 0–10,000



Modeled scenario on public data: IMF PortWatch (AIS estimates of total cargo, not food), UN Comtrade 2023, FAOSTAT 2023. Coverage is modeled, not observed. v0.1 · Miguel Ángel Hernández Aranda · Static PDF export · globe and table interactivity in the .pbix

ALTERNATIVE ROUTES

Observed weekly cargo at ports outside Hormuz · IMF PortWatch AIS estimates of total cargo, not food

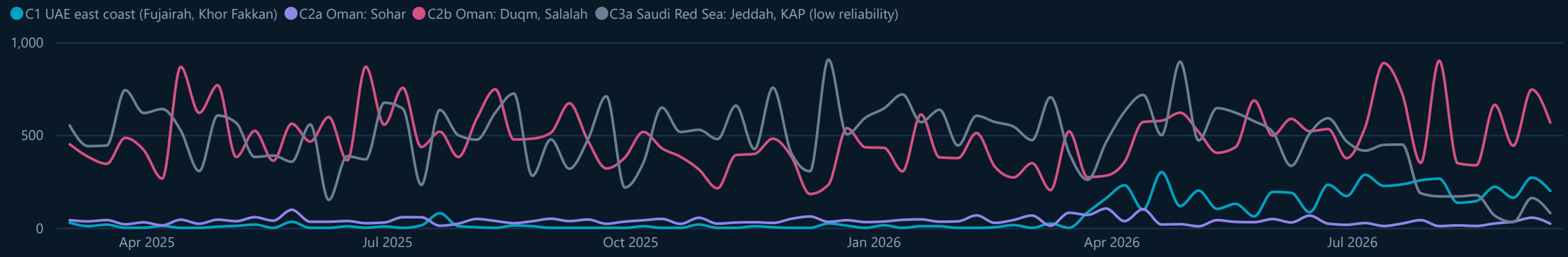
Cargo segment

Container | Dry bulk | General cargo + RoRo

CHOKEPOINT TRANSITS · CARGO SHIPS, NO TANKERS · CLOSURE AVERAGE VS PRE-WAR BASE

-89.6%	-5.1%	+1.9%	-1.0%
Hormuz	Bab el-Mandeb	Suez	Cape of Good Hope

WEEKLY IMPORTS BY CORRIDOR · THOUSAND TONNES PER WEEK · SELECTED SEGMENT



ACTIVITY OVER THE PRE-WAR BASE · T/DAY · C1 TOTAL ROW = BASE + INCREMENT

Scope	Low	Base	High synchronized
C1-only	22,418	28,420	34,674
Regional envelope	29,110	39,899	69,862
C1 total (sensitivity)	23,519	29,521	35,775

ACTIVITY, NOT CAPACITY

PortWatch estimates total cargo from ship signals. It does not show how much of it was food or how much room was left.
Increments over the base count as 0 when a corridor is below it. The C1 total row is total activity: base + increment.
C1-only is the UAE-attributable floor; the regional envelope adds Oman's ports, which also serve Oman's own demand and need a land leg.
C3a (Saudi Red Sea) has a PortWatch anomaly in Aug–Sep 2026 and is only a separate sensitivity.

GAP BY COMMODITY

Gap = requirement × (1 − ACR_ref), split in proportion within each segment · a modeled gap, not an observed shortage

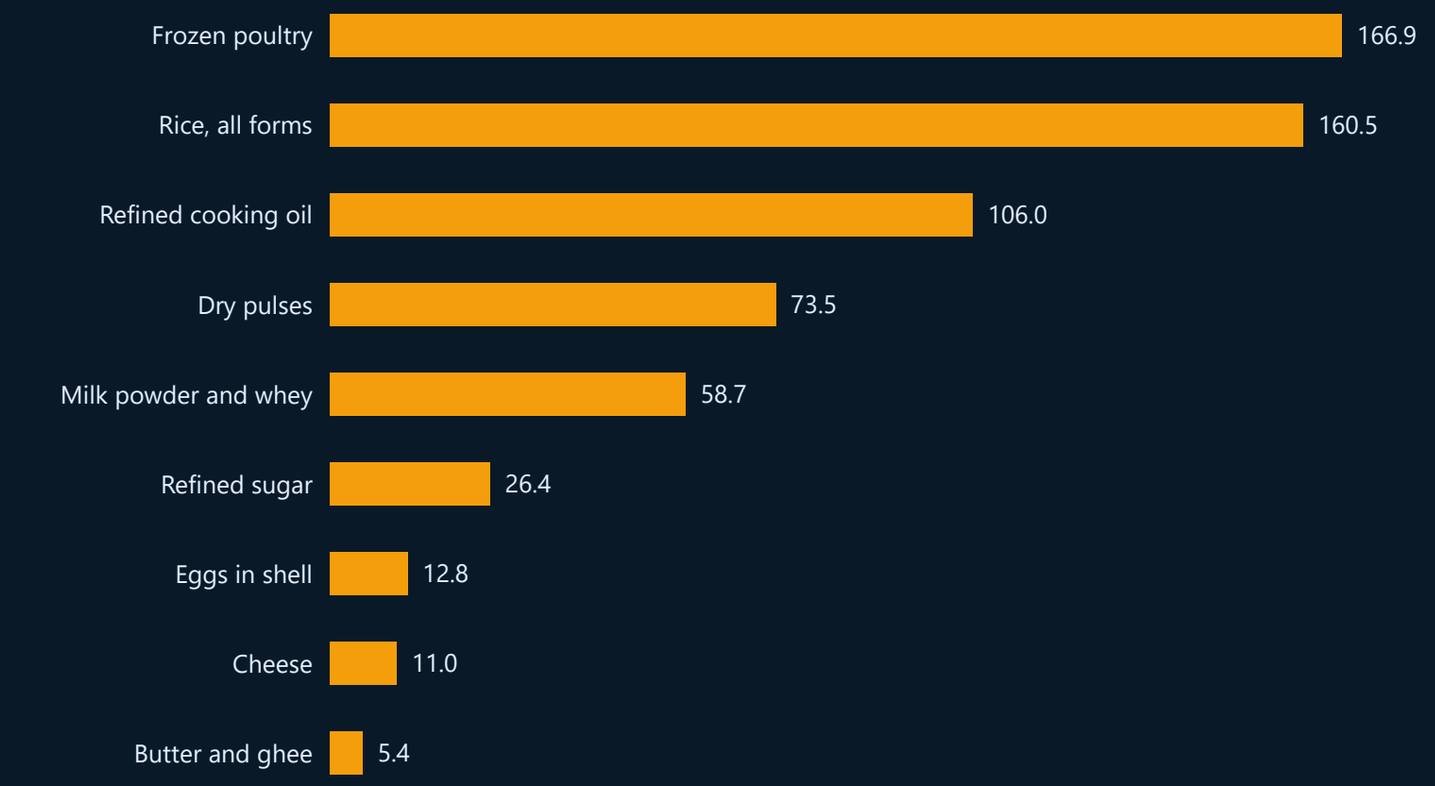
Scope: C1-only | Regional envelope | C1 total (sensitivity)

Activity level: Low | Base | High synchronized

2026 demand: Low | High

Food share of incremental cargo: 19.67% (benchmark)

MODELED GAP BY COMMODITY FORM · T/DAY · PROPORTIONAL ALLOCATION, NOT OBSERVED SHORTAGE



Not shown: forms with a gap ≤0.001 t/day (zero included), forms outside the port model (right) and bread (counted in wheat).

SELECTED SCENARIO · CONTAINERS

87.65%

Container coverage (ACR_ref)

621.3

Container gap · t/day

151

Truck-equivalent trips/day · covered container volume

Lower bound · container forms only · not trips to close the gap

OUTSIDE THE PORT MODEL · T/DAY

202.1

Road: fresh poultry, dairy

138.0

Liquid bulk: crude oil

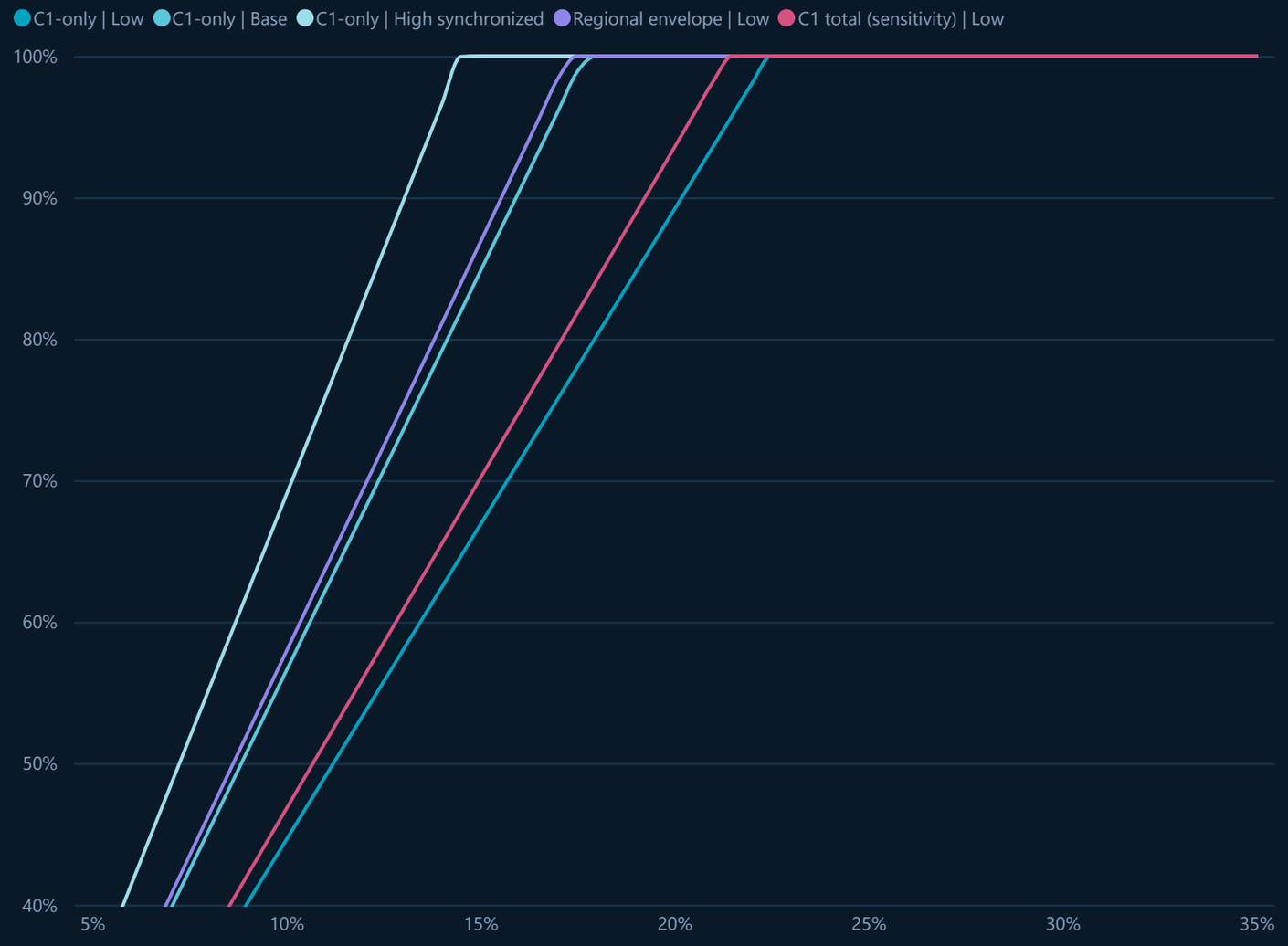
Scenario model on public data: IMF PortWatch (AIS estimates of total cargo, not food), UN Comtrade 2023, FAOSTAT 2023. Modeled coverage, not observed. UAE Food Supply Resilience Stress Test v0.1 · Miguel Ángel Hernández Aranda

FOOD-SHARE SENSITIVITY

Container coverage as the food share of incremental cargo changes (C1 total: total activity, base + increment) · benchmark 19.67%, confidence C

2026 demand: Low | High

CONTAINER ACR_REF BY FOOD SHARE · EACH LINE REACHES 100% AT ITS BREAK-EVEN



FOOD SHARE NEEDED FOR FULL CONTAINER COVERAGE

Scenario	Break-even share
C1-only | Low	22.4%
C1-only | Base	17.7%
C1-only | High synchronized	14.5%
Regional envelope | Low	17.3%
Regional envelope | Base	12.6%
Regional envelope | High synchronized	7.2%
C1 total (sensitivity) | Low	21.4%
C1 total (sensitivity) | Base	17.0%
C1 total (sensitivity) | High synchronized	14.1%

Break-even = requirement ÷ incremental cargo; for the C1 total rows, requirement ÷ total activity (base + increment). A value below the 19.67% benchmark means the scenario is covered under the benchmark.

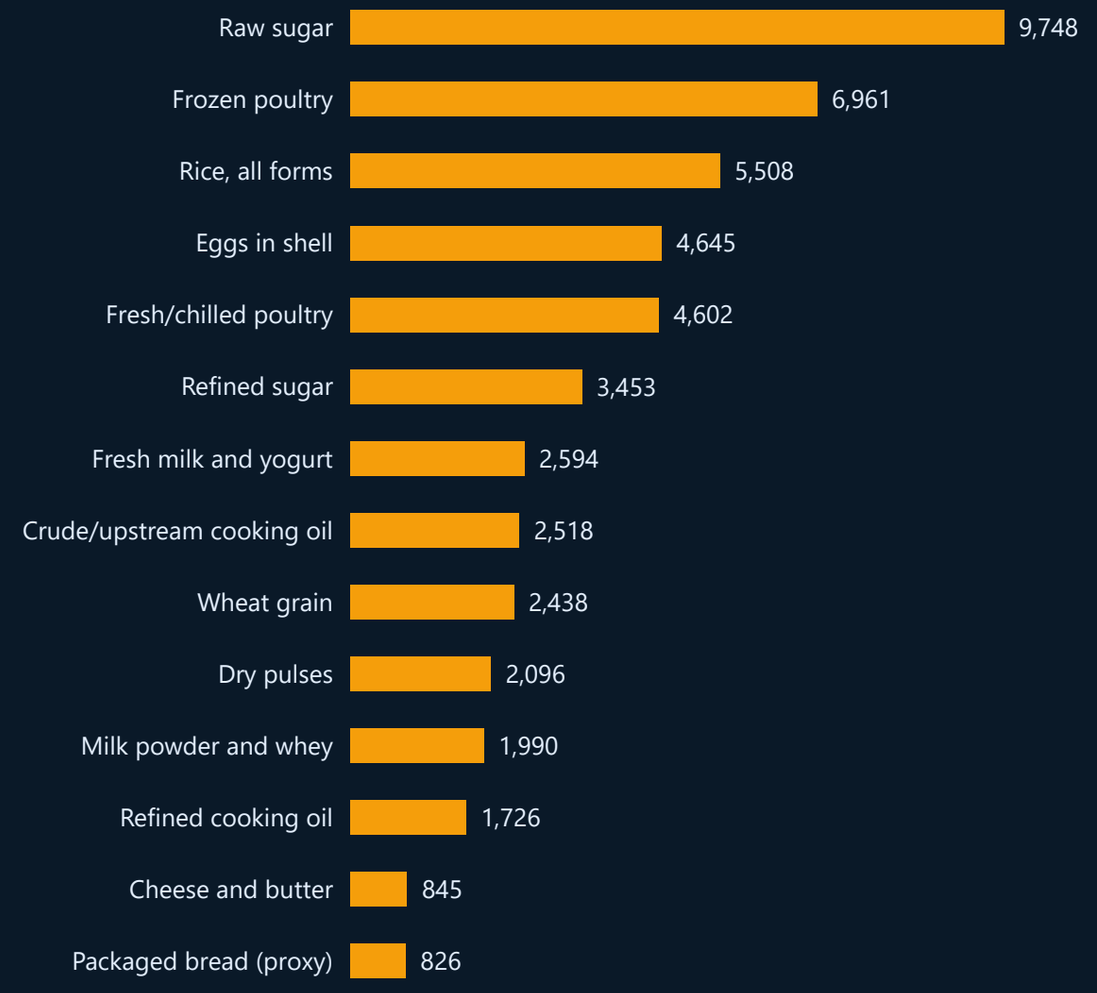
C1 containers · high demand · 19.6656% benchmark: threshold 25,577.5 t/day of total incremental cargo; first 8 weeks 14,852; remaining 21 weeks 25,300.

Scenario model on public data: IMF PortWatch (AIS estimates of total cargo, not food), UN Comtrade 2023, FAOSTAT 2023. Modeled coverage, not observed. UAE Food Supply Resilience Stress Test v0.1 · Miguel Ángel Hernández Aranda

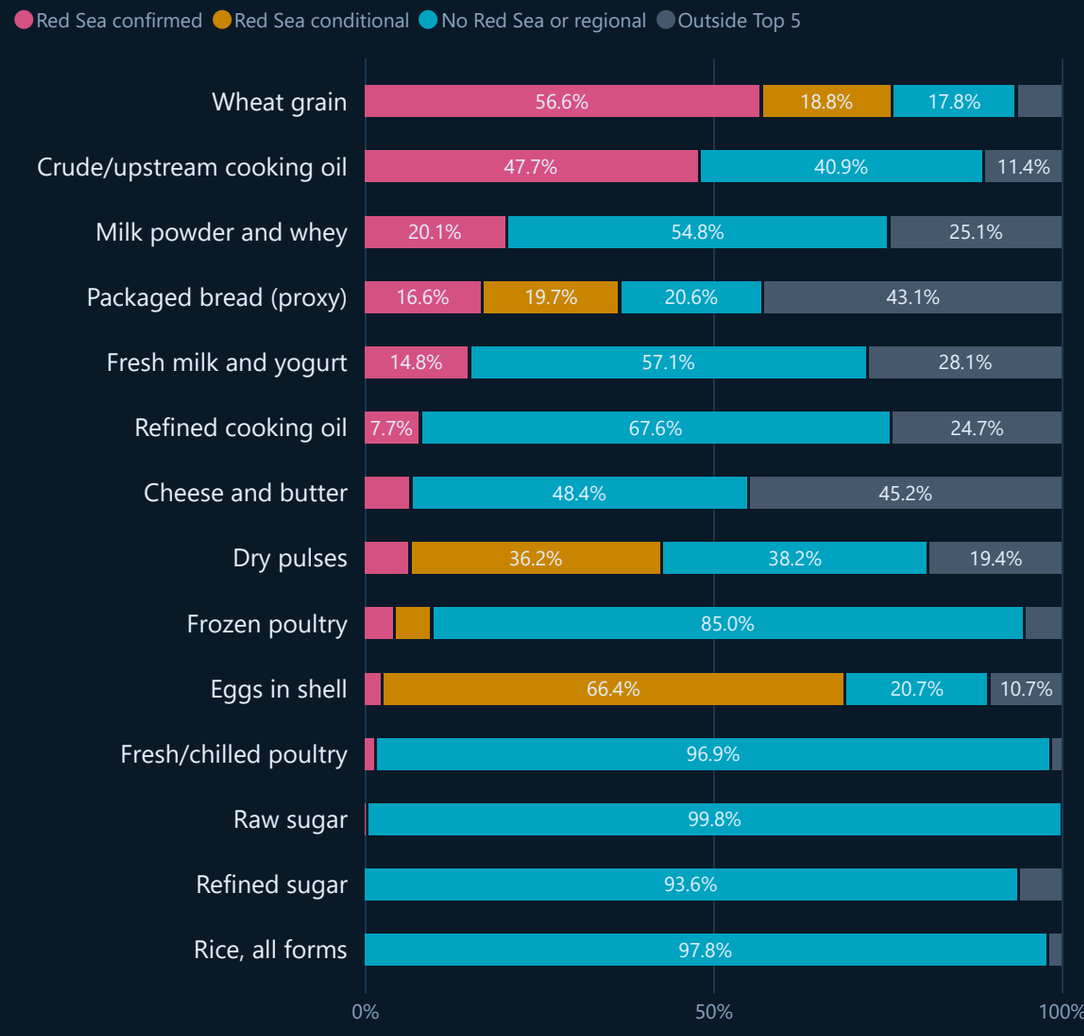
SUPPLIERS AND ROUTES

UN Comtrade 2023 import values · routes inferred by origin country, not tracked by shipment

SUPPLIER CONCENTRATION · HHI 0–10,000



ROUTE EXPOSURE WITHIN THE TOP 5 SUPPLIERS · SHARE OF IMPORT VALUE



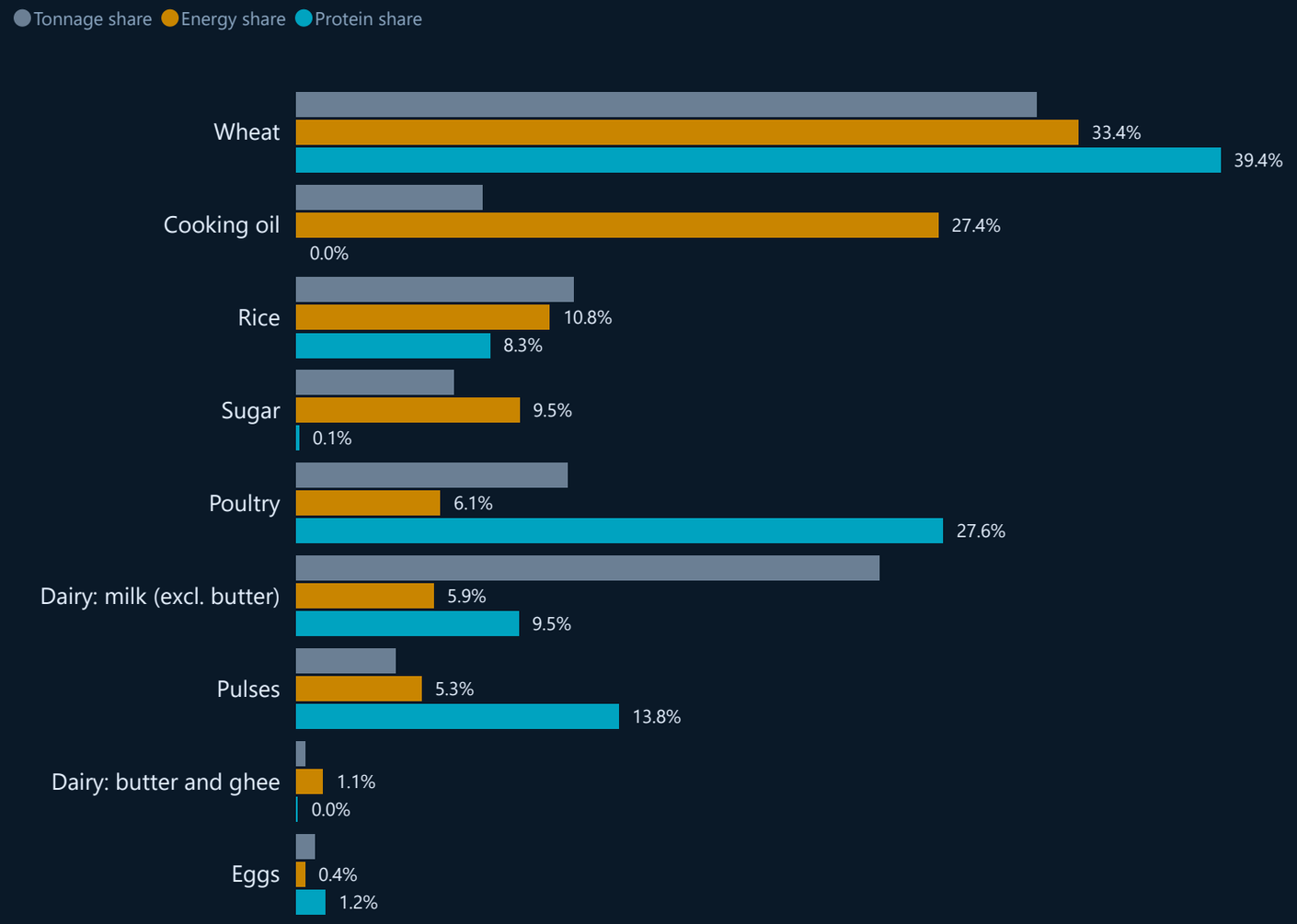
Scenario model on public data: IMF PortWatch (AIS estimates of total cargo, not food), UN Comtrade 2023, FAOSTAT 2023. Modeled coverage, not observed. UAE Food Supply Resilience Stress Test v0.1 · Miguel Ángel Hernández Aranda

NUTRITION

Import-dependent requirement of the analytical basket (2026 residents) by tonnage, energy and protein · FAOSTAT 2023 densities

2026 demand: Low | High

SHARE OF THE IMPORT-DEPENDENT REQUIREMENT BY CATEGORY · LABELS: ENERGY AND PROTEIN · GREY BAR: TONNAGE



IMPORT-DEPENDENT REQUIREMENT OF THE BASKET

32,254.1

Energy · million kcal per day

860.6

Protein · tonnes per day

WHY IT MATTERS

Wheat and cooking oil carry most of the energy; wheat, poultry and pulses most of the protein.
A small disruption in a dense product can matter more than a larger tonnage of another, so priorities by tonnage and by nutrition differ.
Bread is part of wheat and has no separate requirement. Pulses carry low confidence.

METHOD AND LIMITS

Version 0.1 · Results are modeled scenarios based on published trade data, AIS-based activity estimates, inferred routes and stated assumptions.

METHOD

1. Net import requirement: FAOSTAT 2023 food minus counted local production, scaled to 2026 residents (low and high population factors).
2. Shipment weight: requirement × k_ship (Comtrade weight / FAOSTAT imports), then split by physical form before a mode is assigned.
3. Alternative-route activity: PortWatch cargo increment over the pre-war base at C1 (UAE east coast), C2a/C2b (Oman) and C3a (Saudi Red Sea, separate sensitivity). Low = closure average; Base = last 8 weeks; High = best synchronized 4 weeks.
4. Two coverage measures: physical ceiling (ACR_phys) and reference allocation (ACR_ref, allocation share 19.6656%). Gap_ref is a modeled gap, not an observed shortage.
5. Outside the port model: fresh poultry and dairy (TER, regional road route not verified) and crude cooking oil (GL, port coverage not estimable).
6. Risk layers shown beside the gap, never multiplied into tonnes: supplier concentration, inferred route exposure and nutrition.

LIMITS

- PortWatch measures total cargo from ship signals, not food and not capacity.
- The food share of incremental cargo is not observed; the benchmark has confidence C.
- Mode of transport is not observed: regional origin (Saudi Arabia, Oman) does not prove road flow.
- C1 has official road-and-rail continuity evidence; no cross-border rail credit is given to Oman in this version.
- Routes are inferred by origin country; Canada, the USA and eggs from Türkiye are conditional.
- King Abdullah Port shows a data anomaly in Aug–Sep 2026.
- Pulses carry low confidence (large FAOSTAT residual).

SECURITY LIMITS

No reserve stocks, no days-to-depletion, no facility-level capacity, routes or single points of failure. All outputs are at corridor level.

CODE AND DATA

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